

Machine Learning Application Class Environment Setup

1. Install Python

1.1 Windows

- Download Python from the official Python website: <https://www.python.org/downloads/>
- Run the installer.
- Select "Add Python to PATH" before installing. This is important.
- After installation, open Command Prompt or PowerShell and verify the installation:

```
python --version  
pip --version
```

1.2 macOS

- Download Python from the official Python website: <https://www.python.org/downloads/>
- Install Python using the macOS installer.
- After installation, open Terminal and verify the installation:

```
python3 --version  
pip3 --version
```

Note: On macOS, the command is usually python3 instead of python.

2. Create a Project Folder

2.1 Windows

```
mkdir machine-learning-class  
cd machine-learning-class
```

2.2 macOS

```
mkdir machine-learning-class  
cd machine-learning-class
```

3. Create a Virtual Environment

A virtual environment keeps the libraries for this class separate from other Python projects on your computer.

3.1 Windows

```
python -m venv .venv
```

3.2 macOS

```
python3 -m venv .venv
```

4. Activate the Virtual Environment

4.1 Windows - Command Prompt

```
.venv\Scripts\activate
```

4.2 Windows - PowerShell

```
.venv\Scripts\Activate.ps1
```

4.3 macOS

```
source .venv/bin/activate
```

4.4 Confirm Virtual Environment Is Active

The terminal prompt should show (.venv). For example:

```
(.venv) C:\Users\Student\machine-learning-class>
```

5. Install Required Libraries

5.1 Install Core Libraries for This Class

After activating the virtual environment, install the libraries required for the class:

```
pip install numpy pandas matplotlib seaborn scikit-learn jupyter
```

For macOS, if pip does not work, use:

```
pip3 install numpy pandas matplotlib seaborn scikit-learn jupyter
```

5.2 Required Libraries Covered

Library	Purpose
NumPy	Numerical computing and arrays
Pandas	Data loading, cleaning, and analysis
Matplotlib	Basic data visualization and plotting
Seaborn	Statistical data visualization
Scikit-Learn	Classical machine learning models and evaluation
Jupyter Notebook	Interactive notebook environment used during class

6. Install Optional Libraries for Later Classes

The following libraries are not required at the beginning of the class. Install them only when instructed or when they are needed for later topics.

6.1 NLTK

Install NLTK:

```
pip install nltk
```

Download common NLTK datasets:

```
python -m nltk.downloader punkt stopwords wordnet
```

For macOS, if needed:

```
python3 -m nltk.downloader punkt stopwords wordnet
```

6.2 PyTorch

For a normal CPU-only installation, use:

```
pip install torch torchvision torchaudio
```

If your computer has a dedicated NVIDIA GPU and you want GPU support, use the command recommended by the official PyTorch website: <https://pytorch.org/get-started/locally/>

7. Start Jupyter Notebook

Jupyter Notebook is required for the class. Start it from the activated virtual environment:

```
jupyter notebook
```

- A browser window should open automatically.
- Create a new notebook and select the Python environment.
- Keep the terminal open while using Jupyter Notebook.

8. Verify the Installation

Create a new Jupyter Notebook and run the following code cell:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import sklearn

print("NumPy:", np.__version__)
print("Pandas:", pd.__version__)
print("Scikit-Learn:", sklearn.__version__)
print("Setup completed successfully.")
```

Optional verification for later classes:

```
import nltk
import torch

print("NLTK installed successfully.")
print("PyTorch:", torch.__version__)
```

9. Daily Workflow During Class

9.1 Open the Project Folder

Each time you attend class, first open your terminal and go to the project folder:

```
cd machine-learning-class
```

9.2 Activate the Virtual Environment

Windows:

```
.env\Scripts\activate
```

macOS:

```
source .env/bin/activate
```

9.3 Start Jupyter Notebook

```
jupyter notebook
```

9.4 Work on Class Notebooks

- Open the notebook provided by the instructor.
- Run the cells step by step.
- Save your work regularly.
- Keep your terminal open while using Jupyter Notebook.

9.5 End of Class

When finished, close Jupyter Notebook and deactivate the virtual environment:

```
deactivate
```

10. Common Problems and Fixes

10.1 Python Command Not Found

- Windows: reinstall Python and select "Add Python to PATH" during installation.
- macOS: use python3 instead of python.

10.2 pip Command Not Found

Windows:

```
python -m pip install package-name
```

macOS:

```
python3 -m pip install package-name
```

10.3 PowerShell Cannot Activate Virtual Environment

Run this command in PowerShell:

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```

Then activate again:

```
.venv\Scripts\Activate.ps1
```

10.4 Jupyter Notebook Does Not Start

Try:

```
python -m notebook
```

or on macOS:

```
python3 -m notebook
```

If the command still does not work, install the notebook package and try again:

```
pip install notebook  
jupyter notebook
```

10.5 Package Installation Fails

- Check that the virtual environment is activated. The terminal prompt should show (.venv).
- Check your internet connection.
- Try installing the package again.
- Restart the terminal and reactivate the virtual environment.